

Example 5.2

$$\begin{aligned}X_t &= \theta Z_{t-1} + Z_t \\&= (1 + \theta B) Z_t \\&= \alpha(B) Z_t\end{aligned}$$

Where $\alpha(B) = 1 + \theta B$.

$\therefore$ The AGF (autocovariance generating function) is

$$\begin{aligned}\Gamma(z) &= \sigma_z^2 \alpha(z) \alpha(z^{-1}) \\&= \sigma_z^2 \left[(1 + \theta z) \left(1 + \frac{\theta}{z}\right) \right] \\&= \sigma_z^2 \left(1 + \theta z + \frac{\theta}{z} + \theta^2 \right) \\&= \sigma_z^2 \left[(1 + \theta^2) + \theta z + \frac{\theta}{z} \right]\end{aligned}$$

Coefficient of z^0 is $\sigma_z^2 (1 + \theta^2) = \gamma(0)$

Coefficient of z^1 is $\theta \sigma_z^2 = \gamma(1)$

Coefficient of z^{-1} is $\theta \sigma_z^2 = \gamma(-1)$

Coefficient of z^h for $|h| > 1$ is 0

$$\gamma(h) = \begin{cases} \sigma_z^2 (1 + \theta^2) & h = 0; \\ \sigma_z^2 \theta & |h| = 1; \\ 0 & |h| > 1. \end{cases}$$

$$\begin{aligned}
 \gamma(1) &= E(X_t X_{t-1}) \\
 &= E[(\theta z_{t-1} + z_t)(\theta z_{t-2} + z_t)] \\
 &= \theta^2 E(z_{t-1}^2) + \dots
 \end{aligned}$$